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21 July 1980
Superseding
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14 December 1973

MILITARY SPECIFICATION

PLASTIC MOLDING MATERIAL,

EPOXY, GLASS FIBER

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for glass-fiber filled epoxy, transfer, compression, or injection molding materials.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

* Federal

| | |
|---|---|
| PPP-C-96 | Cans, Metal, 28 Gage and Lighter |
| * PPP-P-1892 | Paint, Varnish, Lacquer, and Related Materials, Packaging, Packing and Marking of |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-ED, Redstone Arsenal, AL 35809, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9330

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STANDARDS

Federal

| | |
|---|---|
| FED-STD-406 | Plastics, Methods of Testing |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of US Military Property |
| * MIL-STD-1188 | Commercial Packaging of Supplies and Equipment |

* (Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee

Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, IL).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS



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3.1 Materials. The molding material shall consist of 1/2 inch long chopped fibers of continuous-filament glass, properly impregnated with a precatalyzed, thermo-setting epoxy resin. The resin system shall be an epichlorohydrin bisphenol, a heat-curing epoxy, catalyzed with an amine-type curing agent. After impregnation, the molding material shall be cured up to a moldable state.

3.2 Property values. Standard specimens of the molding material shall conform to the property values shown in Table I. (see 4.8.1)

3.3 Workmanship. The workmanship of the molding material shall be sufficiently high grade to ensure satisfactory performance consistent with the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections of the molding material furnished hereunder shall be classified as follows:

a. Quality conformance inspection. (See 4.4)

b. Packaging inspection. (See 4.5)

4.3 Inspection lot. For purposes of sampling, an inspection lot shall consist of a single batch of molding material offered for inspection at one time. Each batch shall have been produced in one vat in no more than a 24-hour continuous production period with no change in process or raw materials.

4.4 Quality conformance inspection.



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4.4.1 Specimens. The supplier shall submit standard test specimens with each lot of molding material offered for acceptance. The specimens shall have been molded from the batch of material represented, in the form and dimensions specified in Table I. Four specimens shall be submitted for each test.



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TABLE I
Properties of Standard Test Specimens

| Property | Specimen form & dimensions | FED-STD-406 Method | Minimum Value | Unit of Value |
|---|---|---|---|---|
| Arc resistance | 4-inch disc, 1/8 inch thick | 4011 | 125 | Seconds |
| Dielectric constant 1/ at 100 cps | 4-inch disc, 1/8 inch thick | 4021 | 5.5 | --- |
| Dissipation factor 2/ at 100 cps | 4-inch disc, 1/8 inch thick | 4021 | .087 | --- |
| Dielectric strength | 4-inch disc, 1/8 inch thick | 4031 | 360 | Volts/mil |
| Flexural strength | 5-inch bar, 1/4 by 1/2 inch | 1031 | 45,000 | Psi |
| Modulus of elasticity in flexure | 5-inch bar, 1/4 by 1/2 inch | 1031 | $2.5 \times 10^6$ | Psi |
| Compressive strength | 1 by 1/2 by 1/2 inch | 1021 | 33,000 | Psi |
| Modulus of elasticity im compression | 1 by 1/2 by 1/2 inch | 1021 | $1.8 \times 10^6$ | Psi |
| Specific gravity | As specified in FED-STD-406 | 5011 | 1.7 | --- |
| Resin content | As specified in FED-STD-406 | 7061 | 40.0 | Percent |
| Water absorption 24-hour immersion | 2-inch disc, 1/8 inch thick | 7031 | 0.05 | Percent |
| Flammability | 5-inch bar, 1/2 by 1/2 inch | 2021 | 0.05 | Inches/ minute |

1/ Tested in air at room ambient temperature and humidity.
2/ The dissipation factor is the cotangent of the dielectric-phase angle or the tangent of the dielectric-loss angle.

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4.4.2 Testing. Four separate specimens shall be subjected to each of the tests in Table I.

4.4.3 Action in case of failure. One failure shall be allowed in each test. If more than one failure occurs in a single test, the lot represented by the specimens shall be rejected. The supplier shall take corrective action as necessary to preclude recurrence of the failure in subsequently-manufactured lots. Acceptance of subsequently manufactured lots may be withheld until the corrective action has been taken, and four specimens molded from a batch of the molding material which incorporates the corrective action have been subjected to and have passed the quality conformance inspection.

* 4.5 Packaging inspection. Each lot offered for acceptance shall be subjected to the packaging inspection.

* 4.5.1 Sampling and testing. Samples shall be taken in accordance with the applicable specifications and standards referenced in 5.1.1 and 5.1.2. When no sampling plan is specified by those specifications and standards, samples shall be taken in accordance with MIL-STD-105, AQL 2.5. The samples shall be inspected as specified in 4.8.2.

4.5.2 Action in case of failure. Unless other criteria are defined in the applicable documents referenced in 5.1.1 and 5.1.2, failure of one specimen in one or more respects shall be charged as a single defective. If the quantity of defectives exceeds that permitted, the lot represented by the sample shall be rejected. Defective specimens shall not be delivered on contract or purchase order until all deficiencies have been corrected.

4.6 Inspection equipment and facilities. Inspection equipment shall be adequate quality and quantity to permit performance of the required examinations and tests, and shall comply with the equipment described in the applicable inspection methods of this specification and the documents referred to thereby. Inspection facilities shall be adequate to permit performance of the required inspection.

4.7 Standard inspection conditions. Unless otherwise specified, all examinations and tests shall be performed under conditions of room ambient temperature, pressure and humidity.



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4.8 Inspection methods.

4.8.1 Properties tests. Tests shall be conducted in accordance with the applicable test methods of FED-STD-406 to verify compliance with the requirements specified in Table I.

* 4.8.2 Packaging inspection method. Inspection shall be performed to determine whether the molding material is packaged in accordance with the requirements in 5.1.1 and 5.1.2. (See 4.5)

* 5. PACKAGING

* 5.1 Preservation, packing, unitization, and marking. Level A, B or commercial packaging (see 6.2)

* 5.1.1 Level A or B. Preservation, packing, unitization, and marking shall be in accordance with PPP-P-1892 and specifications references therein.

* 5.1.2 Commercial. Preservation, packing, unitization and marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The molding material is intended for use in reinforced structural plastic applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

* b. Level of packaging required. (See 5.1)

* c. Size of container (See PPP-P-1892).

d. Required warranty of shelf life, if any. Terms such as the following are suggested: "The supplier shall warrant that the molding material will meet all requirements hereof following six months of storage in the unopened container at the recommended temperature, and that he will replace any container of molding material which fails to meet all requirements hereof following such term as storage".



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* 6.3 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
PLASTIC MOLDING MATERIAL, EPOXY, GLASS FIBER MIL-P-46892A(MI)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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